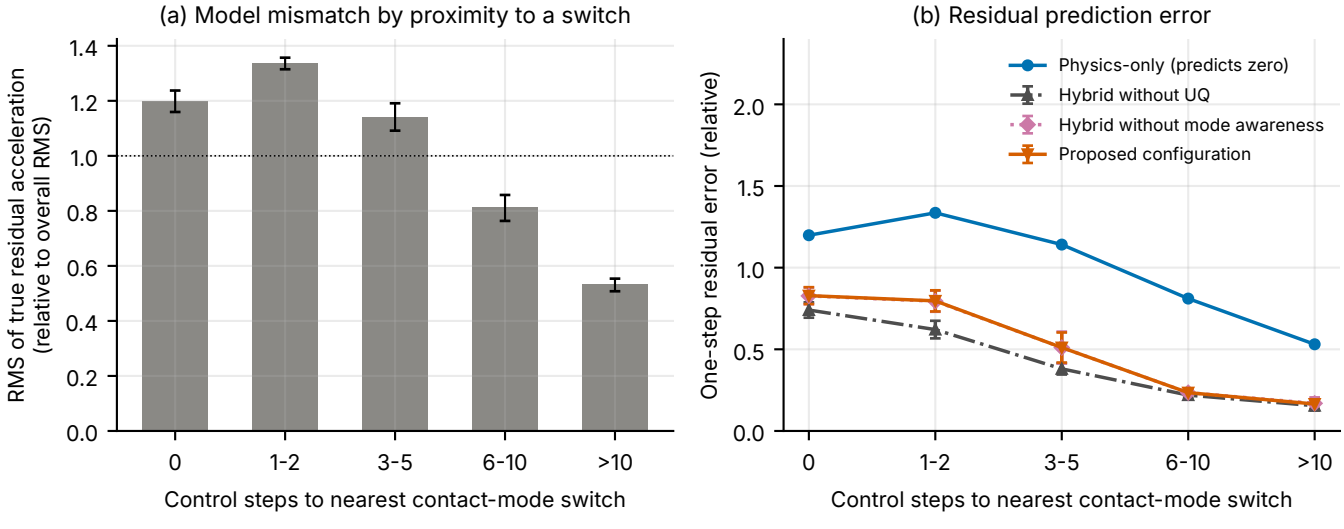
(a) Model mismatch by proximity to a switch
RMS of true residual acceleration (relative to overall RMS)
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
1-2
3-5
6-10
>10
Control steps to nearest contact-mode switch
(b) Residual prediction error
One-step residual error (relative)
2.0
1.5
1.0
0.5
0.0
Physics-only (predicts zero)
Hybrid without UQ
Hybrid without mode awareness
Proposed configuration
0
1-2
3-5
6-10
>10
Control steps to nearest contact-mode switch